Section 4.2

2. To convert from decimal to binary, we successively divide by 2. We write down the remainders so obtained from right to left; that is the binary representation of the given number.

a) Since $321/2$ is 160 with a remainder of 1, the rightmost digit is 1. Then since $160/2$ is 80 with a remainder of 0, the second digit from the right is 0. We continue in this manner, obtaining successive quotients of 40, 20, 10, 5, 2, 1, and 0, and remainders of 0, 0, 0, 0, 1, 0, and 1. Putting all these remainders in order from right to left we obtain $(1\ 0100\ 0001)_2$ as the binary representation. We could, as a check, expand this binary numeral: $2^0 + 2^6 + 2^8 = 1 + 64 + 256 = 321$.

b) We could carry out the same process as in part (a). Alternatively, we might notice that $1023 = 1024 - 1 = 2^{10} - 1$. Therefore the binary representation is 1 less than $(100\ 0000\ 0000)_2$, which is clearly $(11\ 1111\ 1111)_2$.

c) If we carry out the divisions by 2, the quotients are 50316, 25158, 12579, 6289, 3144, 1572, 786, 393, 196, 98, 49, 24, 12, 6, 3, 1, and 0, with remainders of 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, and 1. Putting the remainders in order from right to left we have $(1\ 1000\ 1001\ 0001\ 1000)_2$.

26. In effect, this algorithm computes $11 \mathbf{\ mod\ } 645$, $11^2 \mathbf{\ mod\ } 645$, $11^4 \mathbf{\ mod\ } 645$, $11^8 \mathbf{\ mod\ } 645$, $11^{16} \mathbf{\ mod\ } 645$, $\ldots$, and then multiplies (modulo 645) the required values. Since $644 = (1010000100)_2$, we need to multiply together $11^4 \mathbf{\ mod\ } 645$, $11^{128} \mathbf{\ mod\ } 645$, and $11^{512} \mathbf{\ mod\ } 645$, reducing modulo 645 at each step. We compute by repeatedly squaring: $11^2 \mathbf{\ mod\ } 645 = 121$, $11^4 \mathbf{\ mod\ } 645 = 121^2 \mathbf{\ mod\ } 645 = 14641 \mathbf{\ mod\ } 645 = 451$, $11^8 \mathbf{\ mod\ } 645 = 451^2 \mathbf{\ mod\ } 645 = 203401 \mathbf{\ mod\ } 645 = 226$, $11^{16} \mathbf{\ mod\ } 645 = 226^2 \mathbf{\ mod\ } 645 = 51076 \mathbf{\ mod\ } 645 = 121$. At this point we notice that 121 appeared earlier in our calculation, so we have $11^{32} \mathbf{\ mod\ } 645 = 121^2 \mathbf{\ mod\ } 645 = 451$, $11^{64} \mathbf{\ mod\ } 645 = 451^2 \mathbf{\ mod\ } 645 = 226$, $11^{128} \mathbf{\ mod\ } 645 = 226^2 \mathbf{\ mod\ } 645 = 121$, $11^{256} \mathbf{\ mod\ } 645 = 451$, and $11^{512} \mathbf{\ mod\ } 645 = 226$. Thus our final answer will be the product of 451, 121, and 226, reduced modulo 645. We compute these one at a time: $451 \cdot 121 \mathbf{\ mod\ } 645 = 54571 \mathbf{\ mod\ } 645 = 391$, and $391 \cdot 226 \mathbf{\ mod\ } 645 = 88366 \mathbf{\ mod\ } 645 = 1$. So $11^{644} \mathbf{\ mod\ } 645 = 1$. A computer algebra system will verify this; use the command "`1 &^ 644 mod 645;`" in *Maple*, for example. The ampersand here tells *Maple* to use modular exponentiation, rather than first computing the integer 11^{644}, which has over 600 digits, although it could certainly handle this if asked. The point is that modular exponentiation is much faster and avoids having to deal with such large numbers.

32. The key fact here is that $10 \equiv -1 \pmod{11}$, and so $10^k \equiv (-1)^k \pmod{11}$. Thus 10^k is congruent to 1 if k is even and to -1 if k is odd. Let the decimal expansion of the integer a be given by $(a_{n-1}a_{n-2}\ldots a_3a_2a_1a_0)_{10}$. Thus $a = 10^{n-1}a_{n-1} + 10^{n-2}a_{n-2} + \cdots + 10a_1 + a_0$. Since $10^k \equiv (-1)^k \pmod{11}$, we have $a \equiv \pm a_{n-1} \mp a_{n-2} + \cdots - a_3 + a_2 - a_1 + a_0 \pmod{11}$, where signs alternate and depend on the parity of n. Therefore $a \equiv 0 \pmod{11}$ if and only if $(a_0 + a_2 + a_4 + \cdots) - (a_1 + a_3 + a_5 + \cdots)$, which we obtain by collecting the odd and even indexed terms, is congruent to $0 \pmod{11}$. Since being divisible by 11 is the same as being congruent to $0 \pmod{11}$, we have proved that a positive integer is divisible by 11 if and only if the sum of its decimal digits in even-numbered positions minus the sum of its decimal digits in odd-numbered positions is divisible by 11.

33. Let n be a positive integer, and write its binary expansion

$$n = a_k 2^k + a_{k-1} 2^{k-1} + \cdots + a_2 2^2 + a_1 2^1 + a_0 2^0.$$

Since $2 = 3 - 1$, for each $r = 1, \ldots, k$,

$$2^r = (3-1)^r = 3^r - 3^{r-1} + \cdots + 3(-1)^{r-1} + (-1)^r = 3s_r + (-1)^r$$

for an integer s_r. Then

$$\begin{aligned} n &= a_k(3s_k + (-1)^k) + \cdots + a_2(3s_2 + 1) + a_1(3-1) + a_0 \\ &= 3(s_k a_k + \cdots + s_2 a_2 + a_1) + (-1)^k a_k + \cdots + a_2 - a_1 + a_0 \end{aligned}$$

Thus, n is divisible by 3 if and only if the difference of the sum of the binary digits in even places with the sum of binary digits in odd places is divisible by 3.

4.3

4.a) The prime factorization of 39 is: $39 = 3 \times 13$

4.b) The prime factorization of 81 is: $81 = 3^4$

4.c) Since 101 is a prime number, the prime factorization of 101 is: 101

6 To determine how many zeros are at the end of 100!, we need to count the number of factors of 10 in 100!. Since $10 = 2 \times 5$, and there are always more factors of 2 than factors of 5, the number of zeros is determined by the number of factors of 5.

The number of factors of 5 in 100! is given by those numbers divisible by 5. There are $\lfloor \frac{100}{5} \rfloor$ of them. If it is further divisible by 25, we have one extra factor 5 from it. So we have extra $\lfloor \frac{100}{25} \rfloor$ of them. Since $5^3 = 125$ is larger than 100, we know that there are no other factors of 5.

Hence there are in total $\lfloor \frac{100}{5} \rfloor + \lfloor \frac{100}{25} \rfloor = 20 + 4 = 24$ factors of 5 in 100!. Thus, there are **24** zeros at the end of 100!.

□

11 We will prove that $\log_2 3$ is irrational by assuming the opposite and deriving a contradiction.

Assume $\log_2 3$ is rational. This means we can write $\log_2 3$ as a fraction of two integers. Specifically, there exist p and q integers with $q \neq 0$ such that

$$\log_2 3 = \frac{p}{q}.$$

Using the definition of logarithms, we can rewrite this equation as:

$$2^{\frac{p}{q}} = 3$$

Now, raise both sides to the power of q to eliminate the fraction:

$$2^p = 3^q$$

This equation states that some power of 2 is equal to some power of 3. However, this is a contradiction because powers of 2 and powers of 3 are distinct; no power of 2 can ever equal a power of 3 (since 2 and 3 are distinct prime numbers).

Therefore, the assumption that $\log_2 3$ is rational must be false. Thus, $\log_2 3$ is irrational.

□

12 Consider the following sequence of n consecutive integers:

$$(n+1)! + 2, (n+1)! + 3, \ldots, (n+1)! + (n+1).$$

We claim that each of these numbers is composite.

For each k in the range $2 \leq k \leq n+1$, observe that:

$$(n+1)! + k$$

is divisible by k, because $(n+1)!$ is the product of all integers from 1 to $n+1$, which includes k. Therefore, $(n+1)! + k$ is divisible by k, meaning it has a divisor other than 1 and itself, and thus it is composite.

Since there are n consecutive integers from $(n+1)! + 2$ to $(n+1)! + (n+1)$, all of which are composite, we have proven that for every positive integer n, there are n consecutive composite integers.

□

13 Let $p = 3$, then $p + 2 = 5$ and $p + 4 = 7$, which are all primes. So there are three consecutive odd positive integers that are primes. However, this is the only set of three consecutive odd integers where all are prime. □

32.a) We apply the Euclidean algorithm to find $\gcd(1, 5)$:

$$5 = 1 \cdot 5 + 0$$

Since the remainder is 0, we conclude:

$$\gcd(1,5) = 1$$

32.e) We apply the Euclidean algorithm to find $\gcd(1529, 14038)$.

$$14038 = 1529 \cdot 9 + 277$$
$$1529 = 277 \cdot 5 + 144$$
$$277 = 144 \cdot 1 + 133$$
$$144 = 133 \cdot 1 + 11$$
$$133 = 11 \cdot 12 + 1$$
$$11 = 1 \cdot 11 + 0$$

Since the remainder is now 0, the greatest common divisor is the last nonzero remainder:

$$\gcd(1529, 14038) = 1$$

50 We are given that $a \equiv b \pmod{m}$. This means:

$$a - b = km \quad \text{for some integer } k.$$

Thus, $a = b + km$.

Now, let $d = \gcd(a, m)$. By the definition of the greatest common divisor, d divides both a and m. Since $a = b + km$, it follows that d also divides b (because it divides both a and km). Therefore, d divides both b and m, implying $d \leq \gcd(b, m)$.

Similarly, let $d' = \gcd(b, m)$. Since $a = b + km$ and d' divides both b and m, it follows that d' divides a. Thus, $d' \leq \gcd(a, m)$.

Since $d \leq d'$ and $d' \leq d$, we conclude that:

$$\gcd(a, m) = \gcd(b, m).$$

52 We disprove the statement by finding a counterexample.

Consider $n = 6$, where the first six primes are $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, $p_4 = 7$, $p_5 = 11$, and $p_6 = 13$. Then:

$$p_1p_2p_3p_4p_5p_6 + 1 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30030 + 1 = 30031.$$

Now, check if 30031 is prime. Dividing 30031 by 59, we find:

$$30031 \div 59 = 509 \text{ (exact division).}$$

Thus, 30031 is not prime (it is divisible by 59).

Therefore, the statement is false: $p_1p_2 \ldots p_n + 1$ is not prime for every positive integer n.

□

53 Consider the arithmetic progression $ak + b$, where a and b are positive integers and $k = 1, 2, \ldots$. Suppose that all terms in the arithmetic progression are prime numbers. Then, for each k, $ak + b$ is prime. Let $k = a \cdot b$. Then the corresponding term in the progression is:

$$a(a \cdot b) + b = a^2b + b = b(a^2 + 1).$$

This expression shows that $ak + b = b(a^2 + 1)$ is divisible by b and is larger than b for $a \geq 1$. Since $b(a^2 + 1) > b$, the term is composite because it is greater than b and divisible by b.

Thus, there exists at least one composite number in the arithmetic progression $ak + b$.

□

54 We will prove this by contradiction. Suppose there are only finitely many primes of the form $3k + 2$, and let these primes be $q_1, q_2, \ldots, q_n$.

Consider the number:

$$N = 3q_1q_2 \ldots q_n - 1.$$

Since $N \equiv 2 \pmod{3}$, any prime divisor of N mod 3 must be 1 or 2 and at least one prime divisor mod 3 to be 2.

Now, observe that N is not divisible by any of the primes $q_1, q_2, \ldots, q_n$ because dividing N by any q_i gives a remainder of -1. Therefore, if N is composite, it must have a prime factor of the form

$3k+2$ that is not in the set $\{q_1, q_2, \ldots, q_n\}$, contradicting our assumption that $q_1, q_2, \ldots, q_n$ are the only primes of the form $3k+2$.

Hence, there must be infinitely many primes of the form $3k+2$.

□

4.4

6.a) We use the Extended Euclidean Algorithm to find the inverse of 2 modulo 17. The goal is to find integers x and y such that:

$$2x + 17y = 1.$$

Using the Euclidean algorithm:

$$17 = 8 \cdot 2 + 1$$

Now, work backwards:

$$1 = 17 - 8 \cdot 2.$$

Thus, the inverse of 2 modulo 17 is $x = -8$. Therefore:

$$2^{-1} \equiv -8 \pmod{17}.$$

6.c) Similarly, using the Euclidean algorithm:

$$233 = 1 \cdot 144 + 89$$

$$144 = 1 \cdot 89 + 55$$

$$89 = 1 \cdot 55 + 34$$

$$55 = 1 \cdot 34 + 21$$

$$34 = 1 \cdot 21 + 13$$

$$21 = 1 \cdot 13 + 8$$

$$13 = 1 \cdot 8 + 5$$

$$8 = 1 \cdot 5 + 3$$

$$5 = 1 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1.$$

Now, work backwards to express 1 as a combination of 144 and 233:

$$1 = 3 - 1 \cdot 2$$

$$1 = 3 - 1 \cdot (5 - 1 \cdot 3) = 2 \cdot 3 - 1 \cdot 5$$

$$1 = 2 \cdot (8 - 1 \cdot 5) - 1 \cdot 5 = 2 \cdot 8 - 3 \cdot 5$$

$$1 = 2 \cdot 8 - 3 \cdot (13 - 1 \cdot 8) = 5 \cdot 8 - 3 \cdot 13$$

$$1 = 5 \cdot (21 - 1 \cdot 13) - 3 \cdot 13 = 5 \cdot 21 - 8 \cdot 13$$

$$1 = 5 \cdot 21 - 8 \cdot (34 - 1 \cdot 21) = 13 \cdot 21 - 8 \cdot 34$$

$$1 = 13 \cdot (55 - 1 \cdot 34) - 8 \cdot 34 = 13 \cdot 55 - 21 \cdot 34$$

$$1 = 13 \cdot 55 - 21 \cdot (89 - 1 \cdot 55) = 34 \cdot 55 - 21 \cdot 89$$

$$1 = 34 \cdot (144 - 1 \cdot 89) - 21 \cdot 89 = 34 \cdot 144 - 55 \cdot 89$$

$$1 = 34 \cdot 144 - 55 \cdot (233 - 1 \cdot 144) = 89 \cdot 144 - 55 \cdot 233$$

So the inverse of 144 modulo 233 is 89.

7 Let a and m be relatively prime positive integers, meaning $\gcd(a, m) = 1$. Suppose there are two solutions b and c such that:

$$ab \equiv 1 \pmod{m} \quad \text{and} \quad ac \equiv 1 \pmod{m}.$$

We need to show that $b \equiv c \pmod{m}$.

Subtract the two congruences:

$$ab \equiv ac \pmod{m}.$$

This simplifies to:

$$a(b - c) \equiv 0 \pmod{m}.$$

Since $\gcd(a, m) = 1$, it follows that a has no common factors with m. Therefore, by a property of congruences, $b - c$ must be divisible by m. That is:

$$b - c \equiv 0 \pmod{m},$$

which implies:

$$b \equiv c \pmod{m}.$$

Thus, the inverse of a modulo m is unique modulo m.

$\square$

8 To find the inverse of a modulo m, we need to solve the congruence:

$$ax \equiv 1 \pmod{m},$$

which means there must exist an integer x such that:

$$ax + km = 1.$$

Since $\gcd(a, m)|a$, $\gcd(a, m)|m$, we have $\gcd(a, m)|1$, since $\gcd(a, m)$ is an positive integer, we know $\gcd(a, m) = 1$. Contradict to the assumption that $\gcd(a, m) > 1$.

Thus, no integer x can satisfy the equation, meaning that no inverse of a modulo m exists if $\gcd(a, m) > 1$.

$\square$

12.a) We need to find the inverse of 34 modulo 89. Using the Extended Euclidean Algorithm:

$$\begin{aligned}
89 &= 2 \cdot 34 + 21 \\
34 &= 1 \cdot 21 + 13 \\
21 &= 1 \cdot 13 + 8 \\
13 &= 1 \cdot 8 + 5 \\
8 &= 1 \cdot 5 + 3 \\
5 &= 1 \cdot 3 + 2 \\
3 &= 1 \cdot 2 + 1.
\end{aligned}$$

Now, work backwards to express 1 as a linear combination of 34 and 89:

$$1 = 3 - 1 \cdot 2$$
$$1 = 3 - 1 \cdot (5 - 1 \cdot 3) = 2 \cdot 3 - 1 \cdot 5$$
$$1 = 2 \cdot (8 - 1 \cdot 5) - 1 \cdot 5 = 2 \cdot 8 - 3 \cdot 5$$
$$1 = 2 \cdot 8 - 3 \cdot (13 - 1 \cdot 8) = 5 \cdot 8 - 3 \cdot 13$$
$$1 = 5 \cdot (21 - 1 \cdot 13) - 3 \cdot 13 = 5 \cdot 21 - 8 \cdot 13$$
$$1 = 5 \cdot 21 - 8 \cdot (34 - 1 \cdot 21) = 13 \cdot 21 - 8 \cdot 34$$
$$1 = 13 \cdot (89 - 2 \cdot 34) - 8 \cdot 34 = 13 \cdot 89 - 34 \cdot 34.$$

Thus, $34^{-1} \equiv -34 \equiv 55 \pmod{89}$

Now, multiply both sides of the congruence $34x \equiv 77 \pmod{89}$ by 34^{-1}:

$$34 \cdot 34^{-1}x \equiv 55 \cdot 77 \pmod{89}.$$

This simplifies to:

$$x \equiv 77 \cdot 55 \equiv 52 \pmod{89}.$$

12.b) We are given that $144^{-1} \equiv 89 \pmod{233}$. Multiply both sides of the congruence $144x \equiv 4 \pmod{233}$ by 89:

$$x \equiv 89 \cdot 4 \pmod{233}.$$

Calculating $89 \cdot 4 = 356$, and $356 \mod 233 = 123$.

Thus, the solution is:

$$x \equiv 123 \pmod{233}.$$

16. a) $2 \times 6 \equiv 1 \pmod{11}$, so the inverse of 2 is 6.
$3 \times 4 \equiv 1 \pmod{11}$, so the inverse of 3 is 4.
$5 \times 9 \equiv 1 \pmod{11}$, so the inverse of 5 is 9.
$7 \times 8 \equiv 1 \pmod{11}$, so the inverse of 7 is 8.
Thus, we can pair the integers as follows:

$$(2,6), \quad (3,4), \quad (5,9), \quad (7,8).$$

16.b) First, we write 10!:

$$10! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10.$$

Using the fact that $a \cdot a^{-1} \equiv 1 \pmod{11}$ for each pair of inverses, we simplify the factorial expression modulo 11:

$$10! \equiv (1 \cdot 10) \cdot (2 \cdot 6) \cdot (3 \cdot 4) \cdot (5 \cdot 9) \cdot (7 \cdot 8) \pmod{11}.$$

Each of the pairs cancels out to 1 modulo 11, except for $1 \cdot 10$, which gives:

$$10! \equiv 1 \cdot 10 \cdot 1 \cdot 1 \cdot 1 \equiv 10 \pmod{11}.$$

Since $10 \equiv -1 \pmod{11}$, we conclude that:

$$10! \equiv -1 \pmod{11}.$$

18.a) Since p is prime, every integer a in the range $2, \dots, p-2$ has a unique inverse b in the same range. If $a = b$, then $a^2 \equiv 1 \pmod{p}$, so $p|(a-1)(a+1)$, which means that $a \equiv 1$ or $a \equiv p-1 \pmod{p}$. However, since we are excluding 1 and $p-1$, all other integers form distinct pairs of inverses.

Thus, the integers $2, 3, \dots, p-2$ can be paired as (a,b) where $a \cdot b \equiv 1 \pmod{p}$. Since there are $p-3$ integers in total, there are $(p-3)/2$ pairs of integers.

□

18.b) We now compute $(p-1)!$:

$$(p-1)! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (p-2) \cdot (p-1).$$

By pairing the integers $2, 3, \dots, p-2$, we get:

$$(p-1)! \equiv 1 \cdot (p-1) \pmod{p}.$$

Since $p-1 \equiv -1 \pmod{p}$, we conclude that:

$$(p-1)! \equiv -1 \pmod{p}.$$

18.c) If n is a positive integer such that:

$$(n-1)! \not\equiv -1 \pmod{n},$$

then, by Wilson's Theorem, we can conclude that n is not a prime number.

□

34 By Fermat's Little Theorem, if p is a prime and $\gcd(a,p) = 1$, then:

$$a^{p-1} \equiv 1 \pmod{p}.$$

Here, $p = 41$ and $a = 23$. Since $\gcd(23, 41) = 1$, Fermat's Little Theorem tells us that:

$$23^{40} \equiv 1 \pmod{41}.$$

Now, to compute $23^{1002} \mod 41$, first reduce the exponent 1002 modulo 40:

$$1002 \mod 40 = 1002 - 25 \cdot 40 = 1002 - 1000 = 2.$$

Therefore:

$$23^{1002} \equiv 23^2 \pmod{41}.$$

Next, compute $23^2 \mod 41$:

$$23^2 = 529,$$

and

$$529 \mod 41 = 529 - 12 \cdot 41 = 529 - 492 = 37.$$

Thus:

$$23^{1002} \equiv 37 \pmod{41}.$$

$\square$

40 We want to show that $42 \mid (n^7 - n)$ for any positive integer n. Notice that $42 = 2 \times 3 \times 7$. Therefore, to prove that 42 divides $n^7 - n$, it suffices to show that $n^7 - n$ is divisible by 2, 3, and 7.

Consider $n^7 - n \pmod 2$.

If $n \equiv 0 \pmod 2$, then $n^7 \equiv 0 \pmod 2$, so $n^7 - n \equiv 0 \pmod 2$. If $n \equiv 1 \pmod 2$, then $n^7 \equiv 1 \pmod 2$, so $n^7 - n \equiv 1 - 1 = 0 \pmod 2$.

In either case, $n^7 - n \equiv 0 \pmod 2$, meaning $n^7 - n$ is divisible by 2.

Consider $n^7 - n \pmod 3$.

If $n \equiv 0 \pmod 3$, then $n^7 \equiv 0 \pmod 3$, so $n^7 - n \equiv 0 \pmod 3$. If $n \equiv 1 \pmod 3$, then $n^7 \equiv 1 \pmod 3$, so $n^7 - n \equiv 1 - 1 = 0 \pmod 3$. If $n \equiv 2 \pmod 3$, then $n^7 \equiv 2^7 \equiv 2 \pmod 3$ (since $2^7 = 128 \equiv 2 \pmod 3$), so $n^7 - n \equiv 2 - 2 = 0 \pmod 3$.

In all cases, $n^7 - n \equiv 0 \pmod 3$, meaning $n^7 - n$ is divisible by 3.

Consider $n^7 - n \pmod 7$.

By Fermat's Little Theorem, for any integer n, we know that:

$$n^7 \equiv n \pmod 7.$$

Therefore, $n^7 - n \equiv 0 \pmod 7$, meaning $n^7 - n$ is divisible by 7.

Since $n^7 - n$ is divisible by 2, 3, and 7, it follows that $n^7 - n$ is divisible by $42 = 2 \times 3 \times 7$. Therefore:

$$42 \mid (n^7 - n).$$

Ps: you can also use Fermat's little theorem for mod 2 and mod 3 case. $\square$